CHAPTER 7

Systematic Investment Plan (SIP) & Accumulation

Annuity-Due Formulations, Geometric Progression Derivations, Dollar-Cost Averaging, and Step-Up Dynamics

---

7.1 Definition & The Annuity-Due Model

A Systematic Investment Plan (SIP) represents a disciplined capital accumulation model where an investor commits a constant periodic cash contribution C into a compounding financial asset. Because contributions are conventionally deposited at the **beginning of each periodic interval**, the financial cash flow stream is categorized mathematically as an **Annuity-Due**.

Definition 7.1 (Closed-Form SIP Future Value Equation)

Let periodic contribution be $C > 0$, nominal annual return rate be $r \geq 0$, and investment horizon be t years. With monthly contributions ($m = 12$), the periodic rate is $r_m = r / 12$ and total contribution count is $N = 12 \cdot t$.
The **terminal future value** FV_{SIP} at the end of month N is:

$$FV_{\text{SIP}} = C \cdot (1 + r_m) \cdot \left[\left((1 + r_m)^N - 1\right) / r_m \right]$$

When $r_m = 0$, the equation reduces to total uncompounded principal: $FV = C \cdot N$.

7.2 First-Principles Derivation via Geometric Series

Step-by-Step Mathematical Proof:

Consider the compounded value of each individual monthly contribution at terminal horizon $t = N$:

- Deposit 1 (at start of Month 1, $t = 0$): compounds for N months $\Rightarrow C(1 + r_m)^N$
- Deposit 2 (at start of Month 2, $t = 1$): compounds for $N–1$ months $\Rightarrow C(1 + r_m)^{N-1}$
- Deposit N (at start of Month N, $t = N–1$): compounds for 1 month $\Rightarrow C(1 + r_m)^1$.

The total terminal wealth is the sum of these N compound values:

- $FV = \sum_{k=1}^{N} C\,(1 + r_m)^k = C \sum_{k=1}^{N} q^k$, where common ratio $q = 1 + r_m$.
- Applying the finite geometric summation formula $\sum_{k=1}^{N} q^k = q\left[(q^N - 1) / (q - 1) \right]$:
- Since $q - 1 = (1 + r_m) - 1 = r_m$, substituting gives:
  $FV = C \cdot (1 + r_m) \cdot \left[\left((1 + r_m)^N - 1\right) / r_m \right]$. ■

Theorem 7.1 (Annuity-Due vs. Annuity-Immediate Relation)

An annuity-due investment stream strictly exceeds an annuity-immediate stream by a compound factor of $(1 + r_m)$:
$\boldsymbol{FV_{\text{due}} = FV_{\text{immediate}} \cdot (1 + r_m)}$

Analytical Consequence: Contributing at the start of each month rather than at month-end guarantees an immediate $+r_m$ percentage boost across the entire terminal accumulated wealth.

7.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: a 15-year equity wealth accumulation plan, and an annually stepped-up SIP compounding trajectory.

Example 7.1: 15-Year Mutual Fund SIP Wealth Creation

Problem Statement: An investor commits ***C* = INR 10,000 per month** into an index equity fund generating an expected return of ***R* = 12.00% per annum** for tenure ***t* = 15 years** (N = 180 months). Calculate total principal invested, final maturity corpus, and total compound wealth created.

Step-by-Step Analytical Solution:

Step 1 (Parameter Normalization):

- Monthly contribution: C = INR 10,000
- Periodic rate: $r_m = 0.12 / 12 = 0.0100$ (1.00% per month)
- Total periods: $N = 15 \times 12 = 180$ months.

Step 2 (Geometric Growth Multiplier):

- Base factor: $1 + r_m = 1.0100$
- Compound term: $(1.01)^{180} \approx 5.99580198$
- Bracket term: $(5.99580198 - 1) / 0.01 = 4.99580198 / 0.01 = 499.580198$

Step 3 (Terminal Future Value Calculation):

- $FV_{SIP} = 10{,}000 \times 1.01 \times 499.580198 = 10{,}100 \times 499.580198$
- **Terminal Corpus:** FV_{SIP} = **INR 5,045,760.00**.

Step 4 (Wealth Decomposition):

- Cumulative Principal Contributed: 10,000 × 180 = **INR 1,800,000.00**.
- Net Compound Wealth Generated: 5,045,760.00 – 1,800,000 = **INR 3,245,760.00**.
- *Analytical Insight:* Pure compound interest accounts for **64.33%** of the terminal portfolio, substantiating the exponential dominance of multi-decade systematic accumulation.

Example 7.2: Step-Up SIP (Annual Contribution Escalation)

Problem Statement: To reflect annual salary increments, an investor starts with C_1 **= INR 10,000 per month** in Year 1, and steps up the monthly contribution by **INR 2,000 each subsequent year** (+20% in Year 2, etc.) over a 3-year horizon at nominal return ***R* = 12.00%** (r_m = 0.01). Compute the terminal corpus and compare against a flat SIP.

Step-by-Step Analytical Solution:

Step 1 (Decomposition into 12-Month Block Annuities):
Terminal horizon t = 36 months. 12-month compounding factor: $F_{12} = 1.01 \times [\ ((1.01)^{12} - 1) / 0.01\ ] = 1.01 \times 12.6825 =$ **12.80933**.

Step 2 (Valuation of Year 1 Block — C_1 = INR 10,000):
- Future value at month 12: 10,000 × 12.80933 = INR 128,093.30.
- Compounded for remaining 24 months to month 36: $128{,}093.30 \times (1.01)^{24} = 128{,}093.30 \times 1.269735 =$ **INR 162,644.50**.

Step 3 (Valuation of Year 2 Block — C_2 = INR 12,000):
- Future value at month 24: 12,000 × 12.80933 = INR 153,711.96.
- Compounded for remaining 12 months to month 36: $153{,}711.96 \times (1.01)^{12} = 153{,}711.96 \times 1.126825 =$ **INR 173,206.50**.

Step 4 (Valuation of Year 3 Block — C_3 = INR 14,000):
- Compounded directly to month 36: 14,000 × 12.80933 = **INR 179,330.62**.

Step 5 (Total Terminal Corpus & Comparison):
- $FV_{\text{Step-Up}} = 162{,}644.50 + 173{,}206.50 + 179{,}330.62 =$ **INR 515,181.62**.
- Principal invested: (10,000 × 12) + (12,000 × 12) + (14,000 × 12) = **INR 432,000.00**.
- Standard Flat SIP Corpus (INR 10,000/mo for 36 mos): $10{,}000 \times 1.01 \times [\ ((1.01)^{36} - 1)/0.01\ ] =$ **INR 435,076.00**.
- *Gain:* Stepping up deposits generated **INR 80,105.62 (+18.4%)** in additional terminal corpus.

7.4 Algorithmic Implementation in Python

Below is the complete implementation from models/investment_plans.py, supporting standard annuity-due SIP and step-up annual contribution acceleration:

Listing: models/investment_plans.py — SIP Accumulation Engine

```
def calculate_sip_future_value(monthly_contribution: float, annual_rate: float, tenure_years: float) -> float:
    """
    Computes annuity-due SIP future value: FV = C * (1+r_m) * [((1+r_m)^N - 1) / r_m].
    """
    if monthly_contribution <= 0.0 or annual_rate < 0.0 or tenure_years <= 0.0:
        raise ValueError("SIP parameters must be non-negative.")
    if annual_rate == 0.0:
        return float(monthly_contribution * tenure_years * 12.0)
    monthly_rate = annual_rate / 12.0
    total_months = int(round(tenure_years * 12.0))
    growth_factor = (1.0 + monthly_rate) ** total_months
    return float(monthly_contribution * (1.0 + monthly_rate) * ((growth_factor - 1.0) / monthly_rate))


def calculate_step_up_sip(
    initial_monthly: float, annual_step_up: float, annual_rate: float, tenure_years: float
) -> float:
    """Computes future value of annually increasing SIP contributions."""
    monthly_rate = annual_rate / 12.0
    total_months = int(round(tenure_years * 12.0))
    accumulated_wealth = 0.0
    for month in range(1, total_months + 1):
        year_index = (month - 1) // 12
        current_contribution = initial_monthly + (year_index * annual_step_up)
        months_remaining = total_months - month + 1
        accumulated_wealth += current_contribution * ((1.0 + monthly_rate) ** months_remaining)
    return float(accumulated_wealth)
```

7.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
Annuity Class	Annuity-Due	Deposits executed at the beginning of each compounding period
Accumulation Multiplier	$s\ddot{\square}_{N\|r} = (1 + r_m)\,[\,((1+r_m)^N - 1) / r_m\,]$	Standard actuarial annuity-due future value symbol
Zero-Rate Limit	$\lim_{r \to 0} FV = C \cdot N$	Uncompounded cumulative sum of raw deposits
Convexity / Velocity	$\partial^2(FV)/\partial N^2 > 0$	Accelerating terminal wealth slope during final years
Step-Up Sensitivity	$\partial(FV)/\partial g > 0$	Supercharges capital base before compounding phase transitions